Le rapport

Installation de Zabbix sur Linux :

Étape 1 : Mise à jour du système

activer les droits sudo : usermod -aG sudo addali

Mise à jour du système: sudo apt update

```
addali@UBUNTU22:~$
addali@UBUNTU22:~$ sudo usermod -aG sudo addali
[sudo] password for addali:
addali@UBUNTU22:~$ sudo apt update -y
Hit:1 http://security.ubuntu.com/ubuntu focal-security InRelease
Hit:2 http://tn.archive.ubuntu.com/ubuntu focal InRelease
Hit:3 http://tn.archive.ubuntu.com/ubuntu focal-updates InRelease
Hit:4 http://tn.archive.ubuntu.com/ubuntu focal-backports InRelease
Reading package lists... Done
Building dependency tree
Reading state information... Done
270 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

Étape 2 : Installation de MariaDB (Base de données)

sudo apt install mariadb-server –y

```
addali@UBUNTU22:~$ sudo apt install mariadb-server -y
[sudo] password for addali:
Reading package lists... Done
Building dependency tree
```

Étape 3 : Sécuriser MariaDB

sudo mysql_secure_installation

```
addali@UBUNTU22:~$ sudo mysql_secure_installation
[sudo] password for addali:

NOTE: RUNNING ALL PARTS OF THIS SCRIPT IS RECOMMENDED FOR ALL MariaDB
      SERVERS IN PRODUCTION USE!  PLEASE READ EACH STEP CAREFULLY!
```

Questions posées par `mysql_secure_installation`

1. **Changer le mot de passe root ?**
   Question : `Change the root password? [Y/n]`
2. **Supprimer les utilisateurs anonymes ?**
   Question : `Remove anonymous users? [Y/n]`
3. **Interdire la connexion root à distance ?**
   Question : `Disallow root login remotely? [Y/n]`
4. **Supprimer la base de test ?**
   Question : `Remove test database and access to it? [Y/n`

5 **Recharger les tables de privilèges ?**
Question : Reload privilege tables now?[Y/n]

```
OK, successfully used password, moving on...

Setting the root password ensures that nobody can log into the MariaDB
root user without the proper authorisation.

You already have a root password set, so you can safely answer 'n'.

Change the root password? [Y/n] n
 ... skipping.

By default, a MariaDB installation has an anonymous user, allowing anyone
to log into MariaDB without having to have a user account created for
them.  This is intended only for testing, and to make the installation
go a bit smoother.  You should remove them before moving into a
production environment.

Remove anonymous users? [Y/n] n
 ... skipping.

Normally, root should only be allowed to connect from 'localhost'.  This
ensures that someone cannot guess at the root password from the network.

Disallow root login remotely? [Y/n] n
 ... skipping.

By default, MariaDB comes with a database named 'test' that anyone can
access.  This is also intended only for testing, and should be removed
before moving into a production environment.

Remove test database and access to it? [Y/n] n
 ... skipping.

Reloading the privilege tables will ensure that all changes made so far
will take effect immediately.

Reload privilege tables now? [Y/n] n
 ... skipping.

Cleaning up...

All done!  If you've completed all of the above steps, your MariaDB
installation should now be secure.

Thanks for using MariaDB!
```

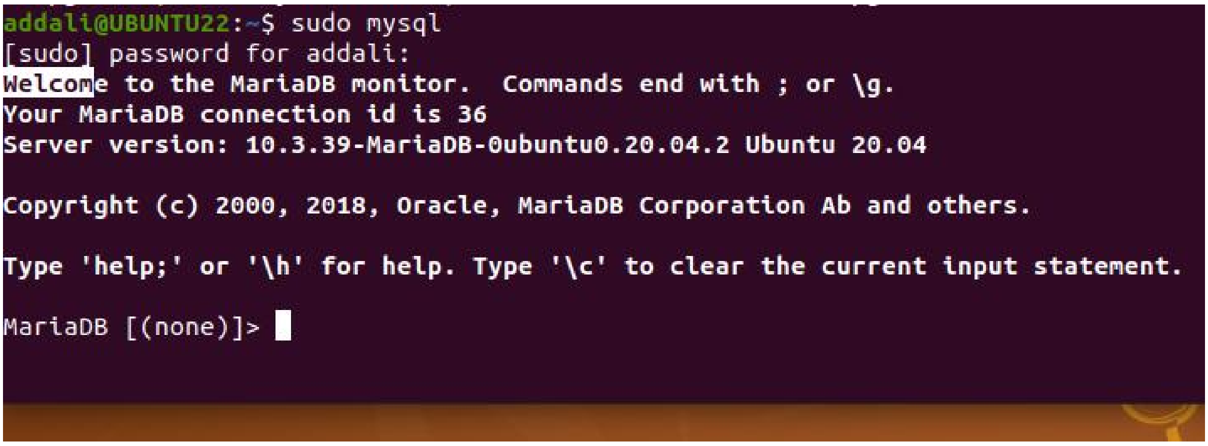
Exmple:

🔐 État de sécurité actuel

Option	Sécurisé ?	Commentaire
Mot de passe root	✅	Déjà défini
Utilisateurs anonymes	✅	Supprimés
Connexion root à distance	❌	Toujours autorisée
Base de test	❌	Toujours présente
Privilèges rechargés	✅	Changements actifs

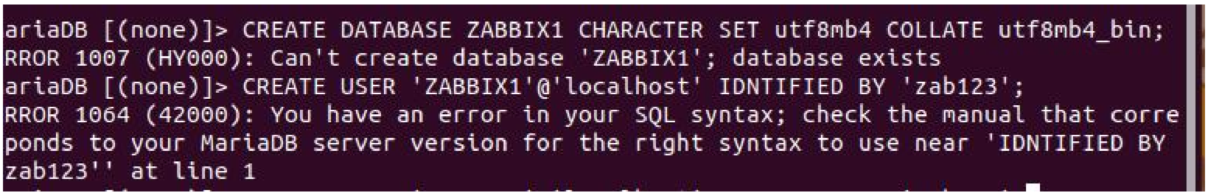
Étape 4 : Création de la base de données Zabbix

sudo mysql



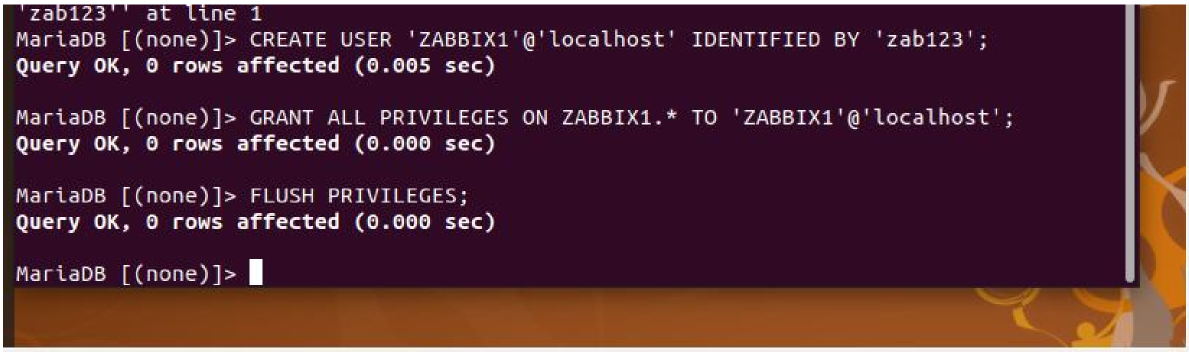
```
addali@UBUNTU22:~$ sudo mysql
[sudo] password for addali:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 36
Server version: 10.3.39-MariaDB-0ubuntu0.20.04.2 Ubuntu 20.04

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]>
```

CREATE DATABASE ZABBIX1 CHARACTER SET utf8mb4 COLLATE utf8mb4_bin;



```
ariaDB [(none)]> CREATE DATABASE ZABBIX1 CHARACTER SET utf8mb4 COLLATE utf8mb4_bin;
RROR 1007 (HY000): Can't create database 'ZABBIX1'; database exists
ariaDB [(none)]> CREATE USER 'ZABBIX1'@'localhost' IDNTIFIED BY 'zab123';
RROR 1064 (42000): You have an error in your SQL syntax; check the manual that corre
ponds to your MariaDB server version for the right syntax to use near 'IDNTIFIED BY
zab123'' at line 1
```

CREATE USER 'ZABBIX1'@'localhost' IDENTIFIED BY 'zab123';
GRANT ALL PRIVILEGES ON ZABBIX1.* TO 'ZABBIX1'@'localhost';
FLUSH PRIVILEGES;
EXIT;



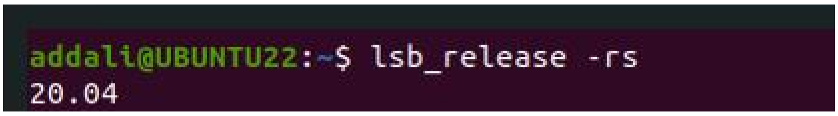
```
'zab123'' at line 1
MariaDB [(none)]> CREATE USER 'ZABBIX1'@'localhost' IDENTIFIED BY 'zab123';
Query OK, 0 rows affected (0.005 sec)

MariaDB [(none)]> GRANT ALL PRIVILEGES ON ZABBIX1.* TO 'ZABBIX1'@'localhost';
Query OK, 0 rows affected (0.000 sec)

MariaDB [(none)]> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.000 sec)

MariaDB [(none)]>
```

Étape 5 :

Ajouter le dépôt officiel de Zabbix

Zabbix n'est pas inclus dans les dépôts officiels d'Ubuntu.

1--Vérifie la version exacte de ton Ubuntu : lsb_release –rs



```
addali@UBUNTU22:~$ lsb_release -rs
20.04
```

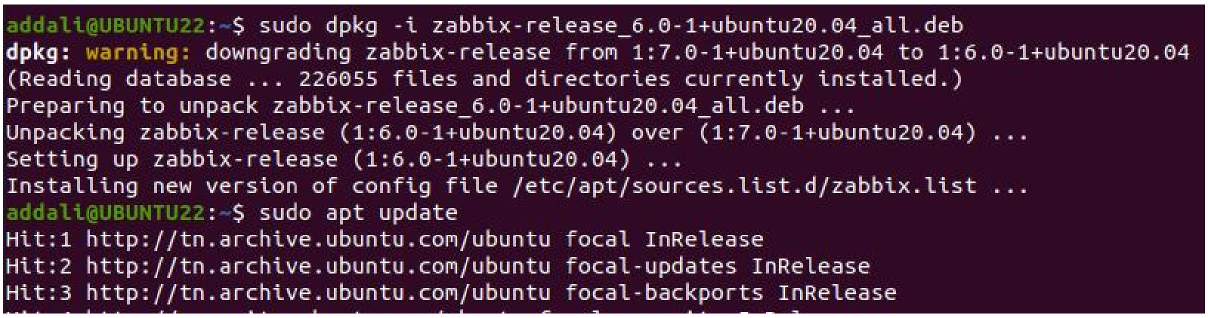
2--Télécharger le paquet release :

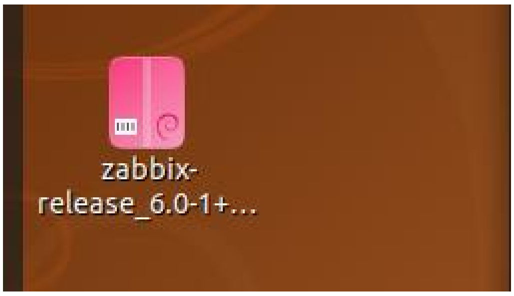
wget https://repo.zabbix.com/zabbix/6.0/ubuntu/pool/main/z/zabbix- release/zabbix-release_6.0-1+ubuntu20.04_all.deb addali@UBUNTU22:~$ wget https://repo.zabbix.com/zabbix/6.0/ubuntu/pool/main/z/zabbix-release/zabbix-release_6.0-1+ubuntu20.04_all.deb

3--Installer le paquet :

sudo dpkg -i zabbix-release_6.0-1+ubuntu20.04_all.deb

```
sudo apt update
```

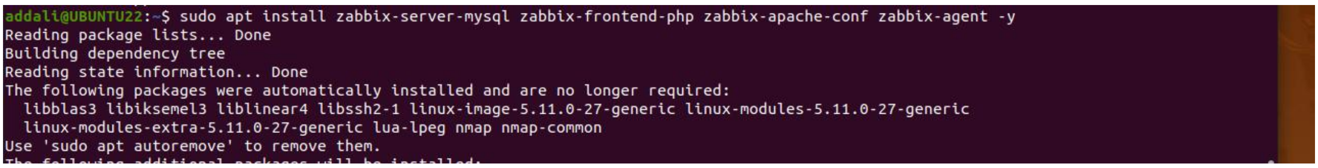




Étape 6 : Installer Zabbix Server + Agent + Interface Web

sudo apt install zabbix-server-mysql zabbix-frontend-php zabbix-apache-conf zabbix-agent –y

```
sudo apt install zabbix-sql-scripts -y
```



. *Vérifier ce qui est installé* `dpkg -l | grep -i zabbix`

```
addali@UBUNTU22:~$ dpkg -l | grep -i zabbix
ii  zabbix-agent                    1:6.0.42-1+ubuntu20.04          amd64
nitoring solution - agent
ii  zabbix-apache-conf              1:6.0.42-1+ubuntu20.04          all
nitoring solution - apache configuration for front-end
ii  zabbix-frontend-php             1:6.0.42-1+ubuntu20.04          all
nitoring solution - PHP front-end
ii  zabbix-release                  1:6.0-1+ubuntu20.04             all
epository configuration
ii  zabbix-server-mysql             1:6.0.42-1+ubuntu20.04          amd64
nitoring solution - server (MySQL)
ii  zabbix-sql-scripts              1:6.0.42-1+ubuntu20.04          all
nitoring solution - sql-scripts
```

Paquet	Version	Architecture	Description simplifiée
zabbix-agent	1:6.0.42-1+ubuntu20.04	amd64	Agent de supervision
zabbix-apache	1:6.0.42-1+ubuntu20.04	all	Configuration Apache pour l'interface web
zabbix-fronter	1:6.0.42-1+ubuntu20.04	all	Interface web en PHP
zabbix-release	1:6.0-1+ubuntu20.04	all	Fichier de configuration du dépôt Zabbix
zabbix-server-	1:6.0.42-1+ubuntu20.04	amd64	Serveur Zabbix avec base MySQL
zabbix-sql-scr	1:6.0.42-1+ubuntu20.04	all	Scripts SQL pour créer la base Zabbix

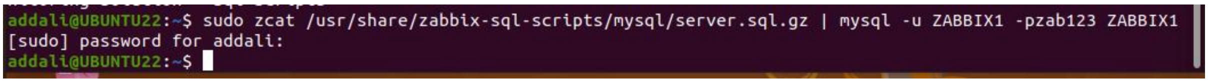
Étape 7 : Importer le schéma de la base Zabbix :

1. Vérifier la version de Zabbix installée: `zabbix_server -V`

```
addali@UBUNTU22:~$ zabbix_server -V
zabbix_server (Zabbix) 6.0.42
Revision 6b0c0216865 29 September 2025, compilation time: Sep 29 2025 13:27:34

Copyright (C) 2025 Zabbix SIA
License GPLv2+: GNU GPL version 2 or later <https://www.gnu.org/licenses/>.
This is free software: you are free to change and redistribute it according to
the license. There is NO WARRANTY, to the extent permitted by law.

This product includes software developed by the OpenSSL Project
for use in the OpenSSL Toolkit (http://www.openssl.org/).

Compiled with OpenSSL 1.1.1f  31 Mar 2020
Running with OpenSSL 1.1.1f  31 Mar 2020
addali@UBUNTU22:~$
```

2.Importer le schéma initial dans la base ZABBIX1 :

sudo zcat /usr/share/zabbix-sql-scripts/mysql/server.sql.gz | mysql -uZABBIX1 -pzab123 ZABBIX1



```
addali@UBUNTU22:~$ sudo zcat /usr/share/zabbix-sql-scripts/mysql/server.sql.gz | mysql -u ZABBIX1 -pzab123 ZABBIX1
[sudo] password for addali:
addali@UBUNTU22:~$
```

Étape 8 :. *Configuration de Zabbix*

Configurer la base de données dans Zabbix

1. *Éditer le fichier de configuration de Zabbix Server* :

sudo nano /etc/zabbix/zabbix_server.conf

```
  GNU nano 4.8                              /etc/zabbix/zabbix_server.comf
DBPassword=zab123
DBHost=localhost
DBName=ZABBIX1
DBUesr=ZABBIX1
```

2.Sauvegarder correctement:

- **CTRL + O** → pour enregistrer
- **Entrée** → pour confirmer
- **CTRL + X** → pour quitter

Étape 9 : Configurer PHP pour Zabbix

1. *la version de PHP* php --

```
addali@UBUNTU22:~$ php --version
PHP 7.4.3-4ubuntu2.29 (cli) (built: Mar 25 2025 18:57:03) ( NTS )
Copyright (c) The PHP Group
Zend Engine v3.4.0, Copyright (c) Zend Technologies
    with Zend OPcache v7.4.3-4ubuntu2.29, Copyright (c), by Zend Technologies
```

version

2. *le fichier de configuration PHP d'Apache pour Zabbix*

sudo nano /etc/zabbix/apache.conf

```
  GNU nano 4.8                    /etc/zabbix/apache.conf                    Modi
# Define /zabbix alias, this is the default
<IfModule mod_alias.c>
    Alias /zabbix /usr/share/zabbix
</IfModule>

<Directory "/usr/share/zabbix">
    Options FollowSymLinks
    AllowOverride None
    Order allow,deny
    Allow from all

    <IfModule mod_php.c>
        php_value max_execution_time 300
        php_value memory_limit 128M
        php_value post_max_size 16M
        php_value upload_max_filesize 2M
        php_value max_input_time 300
        php_value max_input_vars 10000
        php_value always_populate_raw_post_data -1
    </IfModule>
```

créer un fichier PHP ini spécifique : sudo nano /etc/php/7.4/apache2/conf.d/zabbix.ini

```
  GNU nano 4.8                 /etc/php/7.4/apache2/conf.d/zabbix.ini
max_execution-time=300
memory_limit=256M
post_max_size=32M
upload_max_filesize=16M
max_input_time=300
date.timezone=Europe/Paris
max_input_vars=10000
always_populate_raw_post_data=-1
```

ETape10 . CONFIGURATION DU FUSEAU HORAIRE :

1. *Vérifier le fuseau :* timedatectl
2. *Si besoin changer (ex: pour Paris) :* sudo timedatectl set-timezone Europe/Paris

```
addali@UBUNTU22:~$ timedatectl
               Local time: Fri 2025-12-12 12:06:18 WAT
           Universal time: Fri 2025-12-12 11:06:18 UTC
                 RTC time: Fri 2025-12-12 11:06:11
                Time zone: Africa/Lagos (WAT, +0100)
System clock synchronized: yes
              NTP service: active
          RTC in local TZ: no
addali@UBUNTU22:~$ sudo timedatectl set-timezone Europe/Paris
[sudo] password for addali:
addali@UBUNTU22:~$ timedatectl
               Local time: Fri 2025-12-12 12:07:53 CET
           Universal time: Fri 2025-12-12 11:07:53 UTC
                 RTC time: Fri 2025-12-12 11:07:53
                Time zone: Europe/Paris (CET, +0100)
System clock synchronized: yes
              NTP service: active
          RTC in local TZ: no
addali@UBUNTU22:~$ S
```

ÉTAPE 11 : REDÉMARRER LES SERVICES

1. *Vérifier le nom correct du service:*

   `ls /lib/systemd/system/ | grep zabbix`

2. *Activer les services au démarrage :*

   sudo systemctl enable zabbix-server zabbix-agent apache2 mariadb

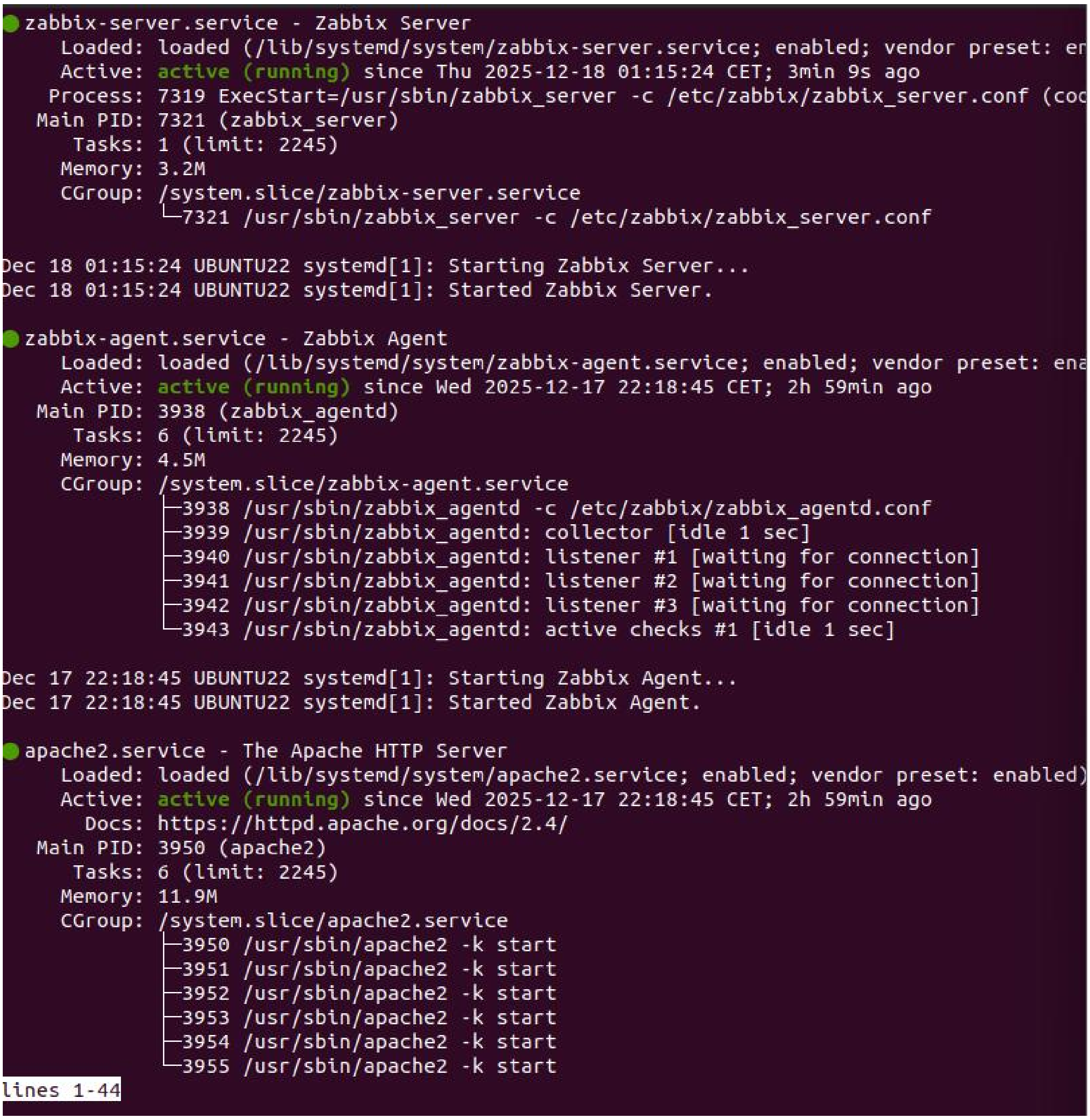
3. *Redémarrer tous les services pour appliquer les changements :*

   sudo systemctl restart zabbix-server zabbix-agent apache2 mariadb

4. *vérifier le statut de chaque service* : *sudo systemctl status zabbix-server*

5. *vérifier tous les services d'un coup :*

`sudo systemctl status zabbix-server zabbix-agent apache2 mariadb`



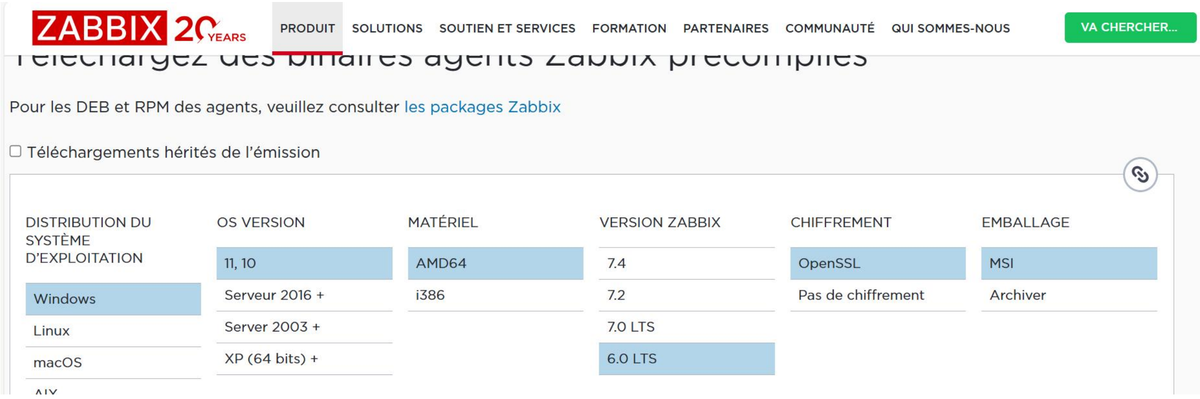
Étape12 Installation de l'Agent Windows pour Zabbix 6.0

1.Télécharger la bonne version :

URL pour Zabbix Agent 6.0 :

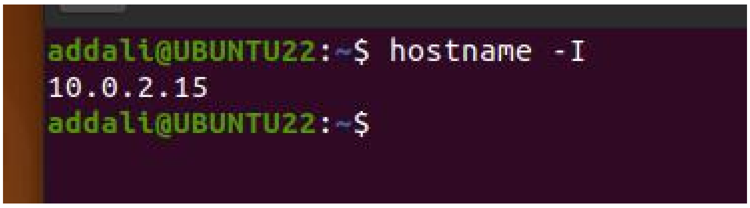
https://www.zabbix.com/download_agents?version=6.0&platform=windows

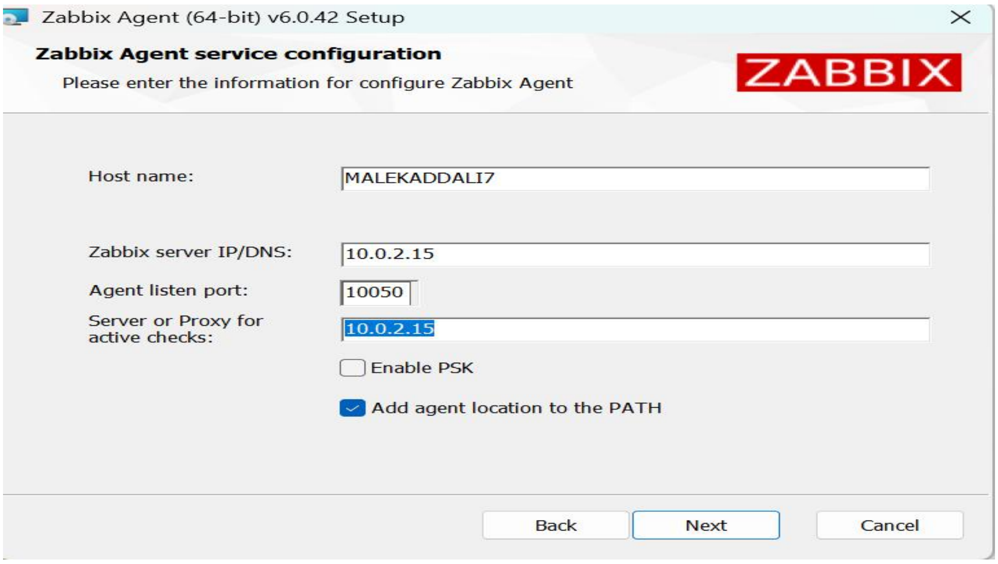
Version spécifique : zabbix_agent-6.0.0-windows-amd64-openssl.msi



2.l'installation :

l'adresse IP de votre serveur Zabbix sous Ubuntu : `hostname -I`

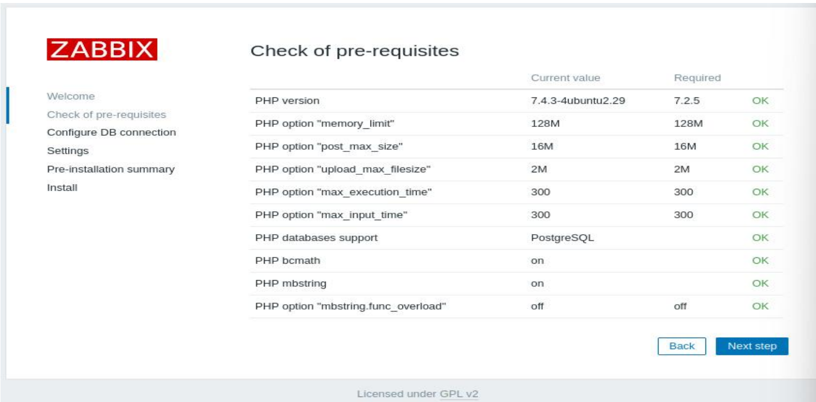




Étape 13 : Finalisation via l'Interface Web :

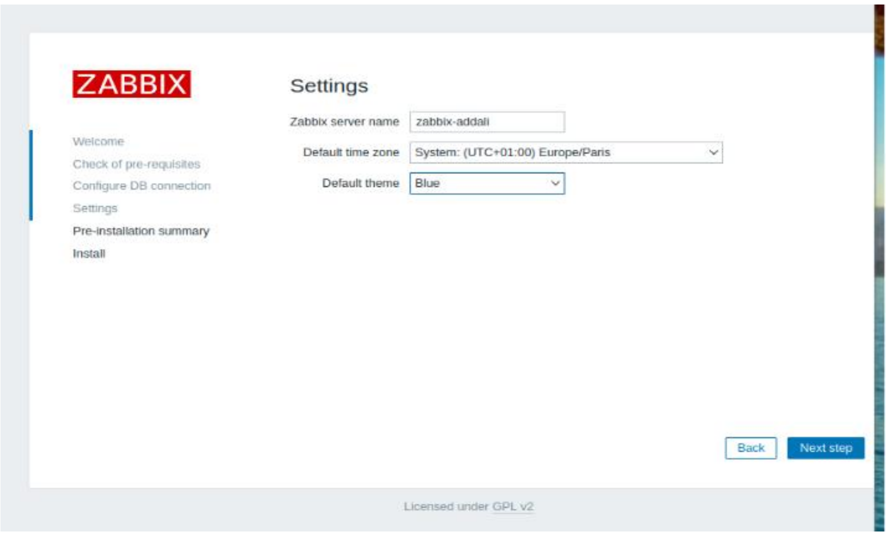
1.Ouvrez votre navigateur et tapez : `http://10.0.2.15/zabbix`

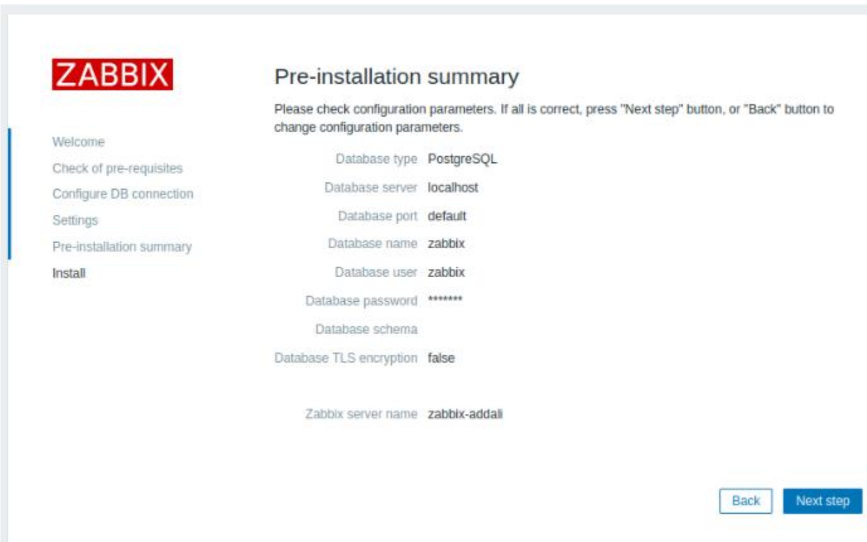
2.Vérifiez que tous les prérequis sont marqués **OK** en vert :



coche la case "TLS encryption" **si** tu as configuré SSL dans PostgreSQL

Si Tu utilises **PostgreSQL en local** (`localhost`), sans configuration SSL/TLS. **Ne coche pas**

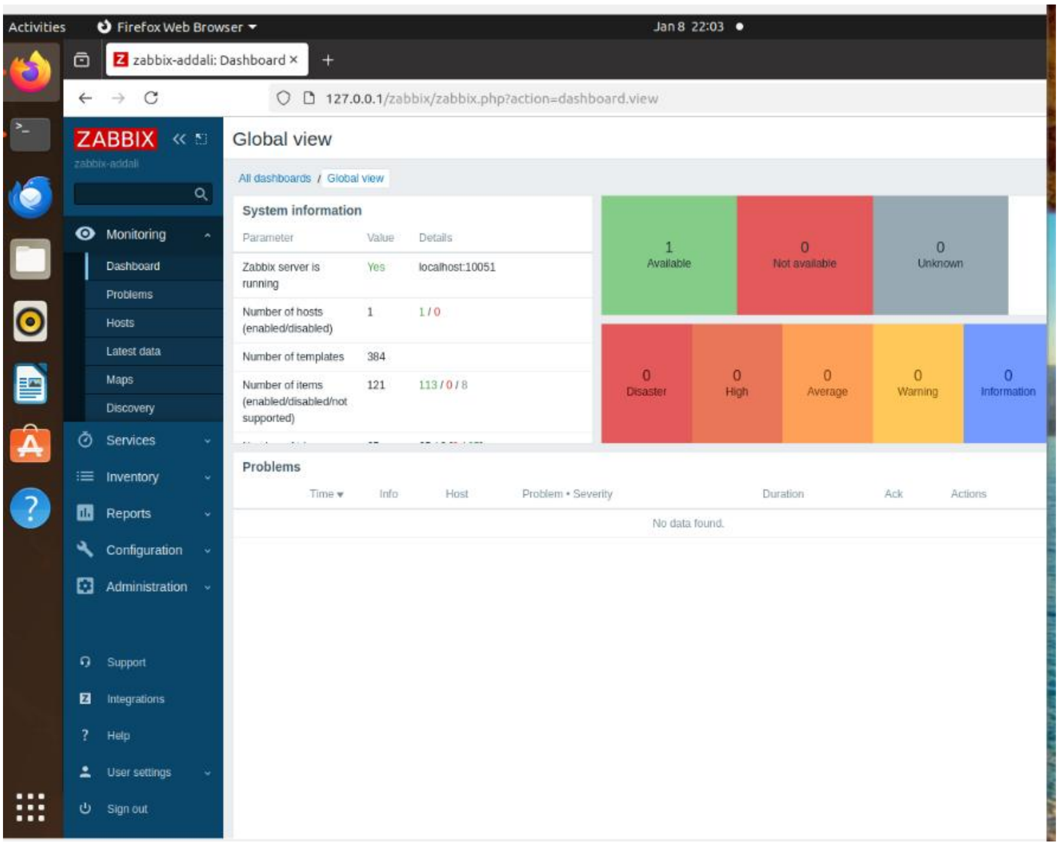




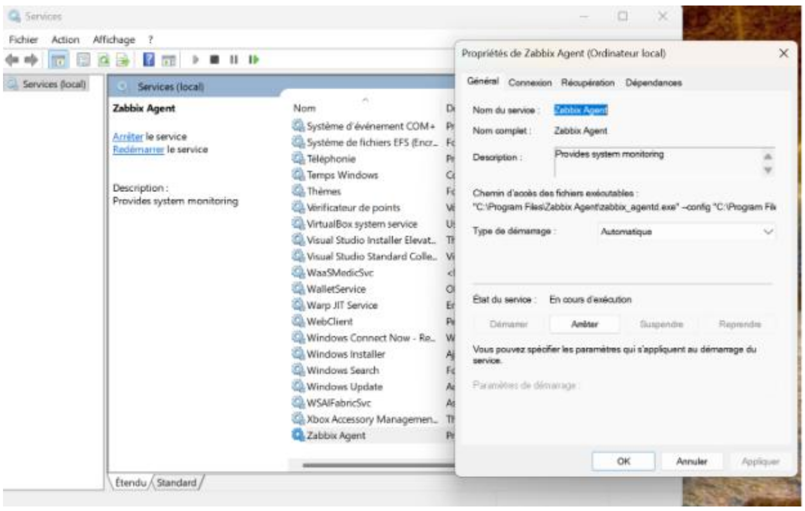
**par défaut :

Username : `Admin` (avec un **A majuscule**).

Password : `zabbix` Le mot de passe initial est **`zabbix`** (tout en minuscule)



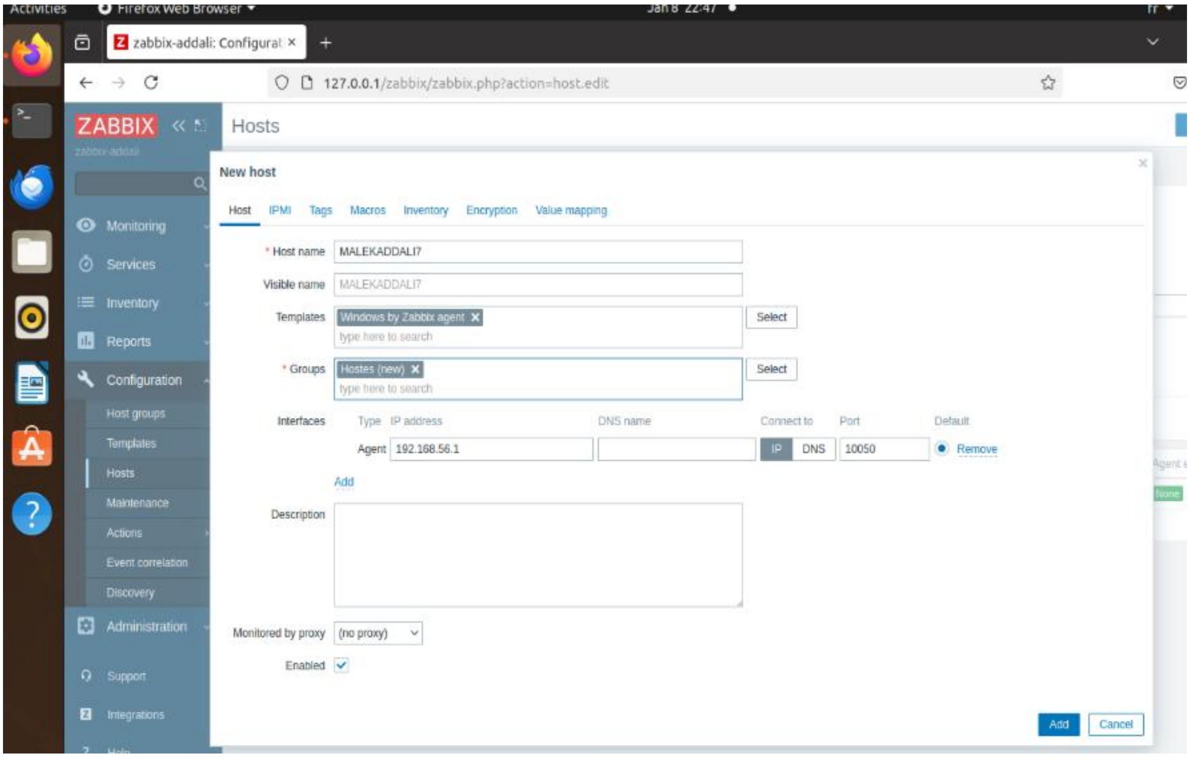
**Vérification de l'état de l'agent Zabbi

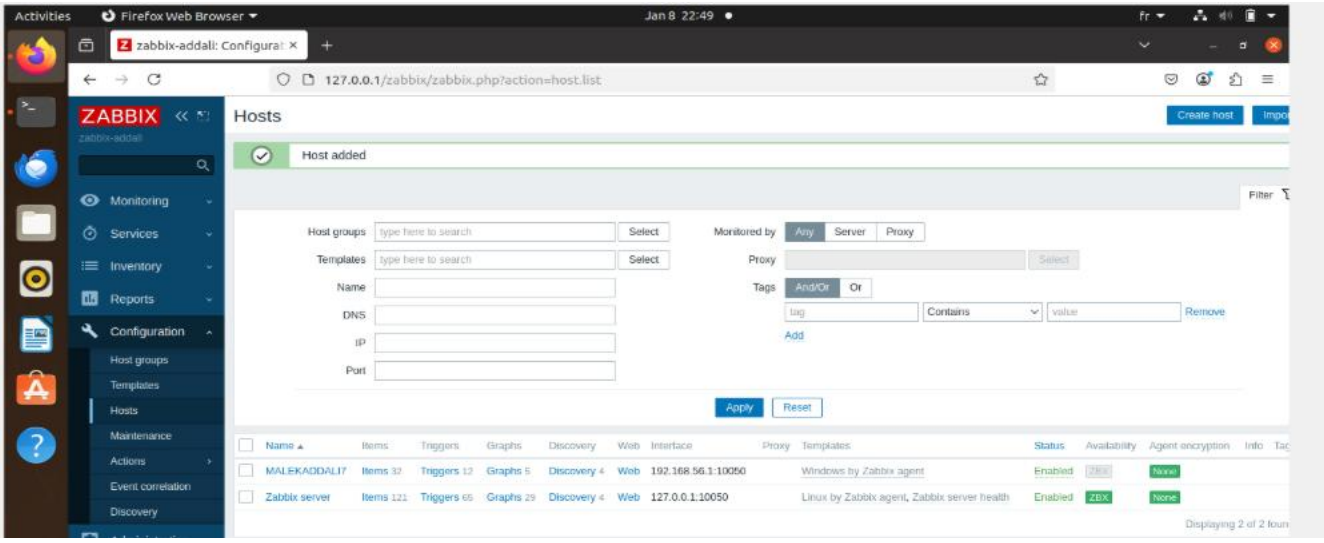


Adresse IP de l'agent : `10.0.2.15`

```
C:\Users\abeda>ipconfig

Configuration IP de Windows


Carte Ethernet Ethernet 3 :

   Suffixe DNS propre à la connexion. . . :
   Adresse IPv6 de liaison locale. . . . .: fe80::fbca:bb71:b0f0:471a%21
   Adresse IPv4. . . . . . . . . . . . . .: 192.168.56.1
   Masque de sous-réseau. . . . . . . . . : 255.255.255.0
   Passerelle par défaut. . . . . . . . . :
```

- **Créer l'hôte dans Zabbix**
  - Va dans **Configuration** → **Hosts** → **Create host**
  - Mets le même **Hostname** que celui défini dans ton agent (`MALEKADDALI7`)
  - **Groupe** : `Hostes (new)`
  - **template** : Windows by Zabbix agent x
  - l'**interface agent** : Add--> IP (`192.168.56.1`) et port `10050`






```
┌─────────────────────────────────────────────┐
│              SERVEUR ZABBIX (Ubuntu)        │
│  • MYSQL (base de données)             │
│  • Zabbix Server + Web Interface            │
│  • Zabbix Agent (auto-monitoring)           │
│                                             │
│  ┌──────────────────────────────────────┐   │
│  │         HÔTES SURVEILLÉS              │    │
│  │ 1. MALECADDALT (Windows)             │    │
│  │     - IP: 192.168.95.1               │    │
│  │     - Port: 10050                    │    │
│  │     - Template: Windows by Zabbix    │    │
│  │                                      │    │
│  │ 2. Zabbix server (localhost)         │    │
│  │     - IP: 127.0.0.1                  │    │
│  │     - Port: 10050                    │    │
```

```
│   │     - Template: Linux by Zabbix        │     │
│   └────────────────────────────────────────┘     │
└──────────────────────────────────────────────────┘
```

Test rapide :

*****Le port 10050 est-il ouvert sur Windows ?**

Depuis le serveur Ubuntu, lance : ping 192.168.56.1

Ouvre PowerShell en admin et tape : netstat -an | findstr :10050

Pour ubuntu : sudo netstat -anp | grep :10050

Cela te montrera si le serveur Ubuntu écoute sur le port `10050`, et si une connexion est établie avec l'agent Windows.

****vérifier si Zabbix Agent sur Windows répond bien au serveur Zabbix sur Ubuntu.**

Installe l'outil zabbix-get sur ubuntu : sudo apt install zabbix-get

Apres: zabbix_get -s 192.168.56.1 -p 10050 -k "system.hostname"

*****Solution : modifier le fichier sur Windows**

Ouvre Le fichier (zabbix_agentd.conf) et vérifie ces lignes :

Server=10.0.2.15

ServerActive=10.0.2.15

Hostname=MALEKADDALI7

Redémarre le service : Restart-Service -Name "Zabbix Agent"

LES SOURCES :

https://youtu.be/nFulGCNc89A?si=j7k_esY7nidQRIou
https://youtu.be/3lc_EEbJhAU?si=VoUCndS1WnqG7RNv